Tectonophysics, 156 (1988) 75–88
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

Paleomagnetism of Permo-Triassic red beds in the western Pyrenees: evidence for strong clockwise rotations of the Paleozoic units

J.J. SCHOTT and A. PERES

Institut de Physique du Globe de Strasbourg, 5 rue Descartes, 67084 Strasbourg Cedex (France)

(Received April 22, 1987; revised version accepted March 28, 1988)

Abstract

Schott, J.J. and Peres, A., 1988. Paleomagnetism of Permo-Triassic red beds in the western Pyrenees: evidence for strong clockwise rotations of the Paleozoic units. *Tectonophysics*, 156: 75–88.

The paleomagnetic study of Permo-Triassic red bed outcrops located in several basement units and slices of the western Pyrenees is presented. Despite strong overprintings attributed to remagnetization processes connected with the early Tertiary compressional phase, we were able to identify the original magnetization in all sites. The directions yielded reveal the occurrence, in the whole area, of clockwise rotations, varying, with respect to the European reference direction, from nearly 0° to 102°. The secondary directions are interpreted as more or less contemporary with the inferred rotations. The paleomagnetic data are discussed in the light of two kinematic models for the Iberian plate. The examination of the models involving a left-lateral displacement of Iberia during the mid-Cretaceous is most conclusive and leads us to suggest that, at that time, the boundary between Europe and Iberia was located south of the Cinco-Villas and Aldudes massifs. The clockwise rotations were induced by the early Tertiary convergence of the plates, which markedly disturbed the rather simple pattern inferred for the mid-Cretaceous period.

1. Introduction

In a model originally proposed by Le Pichon et al. (1971), later modified by Choukroune (1976), and improved by Olivet et al. (1984), the main movement of the Iberian plate relative to Europe occurred in the continental area along a narrow transform zone whose most conspicuous present-day trace would be the North Pyrenean Fault (N.P.F). The existence and width of the transform zone and the role played by the N.P.F. is still a matter of debate (e.g., Boillot and Capdevilla, 1977; Souquet et al., 1977; Bessière and Soula, 1978; Choukroune and Mattauer, 1978). Even if this model is adopted, delineating the boundary between Europe and Iberia remains an unsolved problem because the N.P.F. is identified only in the central Pyrenees. Its western extension is particularly uncertain and gives rise to the question of the significance of the Paleozoic Basque massifs in connection with the Pyrenean axial zone, and of their location relative to either plate. Paleomagnetism can help to solve this problem by delineating the boundary between areas displaying European paleomagnetic directions and areas revealing Iberian directions. Van der Voo (1969) and Van der Voo and Boessenkool (1973) made important contributions showing that European-like directions occur in the Cinco-Villas massif, whereas in the Oroz-Betelu massif the directions resemble the Iberian reference direction. The purpose of this study was to extend investigations to other base-

0040-1951/88/$03.50 © 1988 Elsevier Science Publishers B.V.

ment units recognized in this area, in order to give a more detailed pattern of the paleomagnetic directions.

2. Geology and sampling (Fig. 1)

2.1. Stratigraphy

The sedimentary cycle following the Variscan orogeny was controlled by two narrow basins. One is located in the La Rhune massif, the second—20 km eastwards—is called the Bidarray basin (Soula et al., 1979). The lowermost deposits from the La Rhune massif consist of fossiliferous conglomerates, sandstones and carbonaceous black shales, attributed to the Stephanian (Herbert, 1881; Doubinger and Mouline, 1961). The Permian (Saxonian?, after Lucas, 1977, 1985) is represented in the La Rhune massif, around the Cinco-Villas massif and mainly in the basin of Bidarray where its estimated thickness is about 1000 m. An important volcanic episode took place at the top of the Permian sequence, as indicated by the occurrence of basaltic flows intercalated in the uppermost levels (Bixel and Lucas, 1983). These are unconformably overlain by conglomerates or sandstones representing the beginning of the Triassic red beds which reach a thickness of 200–300 m in the Bidarray basin. A dating based on pollen yields an age as recent as Ladinian–Carnian–Norian (Lucas et al., 1980). The discrepancy between this data and the age of the subsequent marine carbonates attributed to the Ladinian, is discussed by Lucas (1985). As pointed out by this author this relatively young age shows that the continental deposits, classically attributed to the

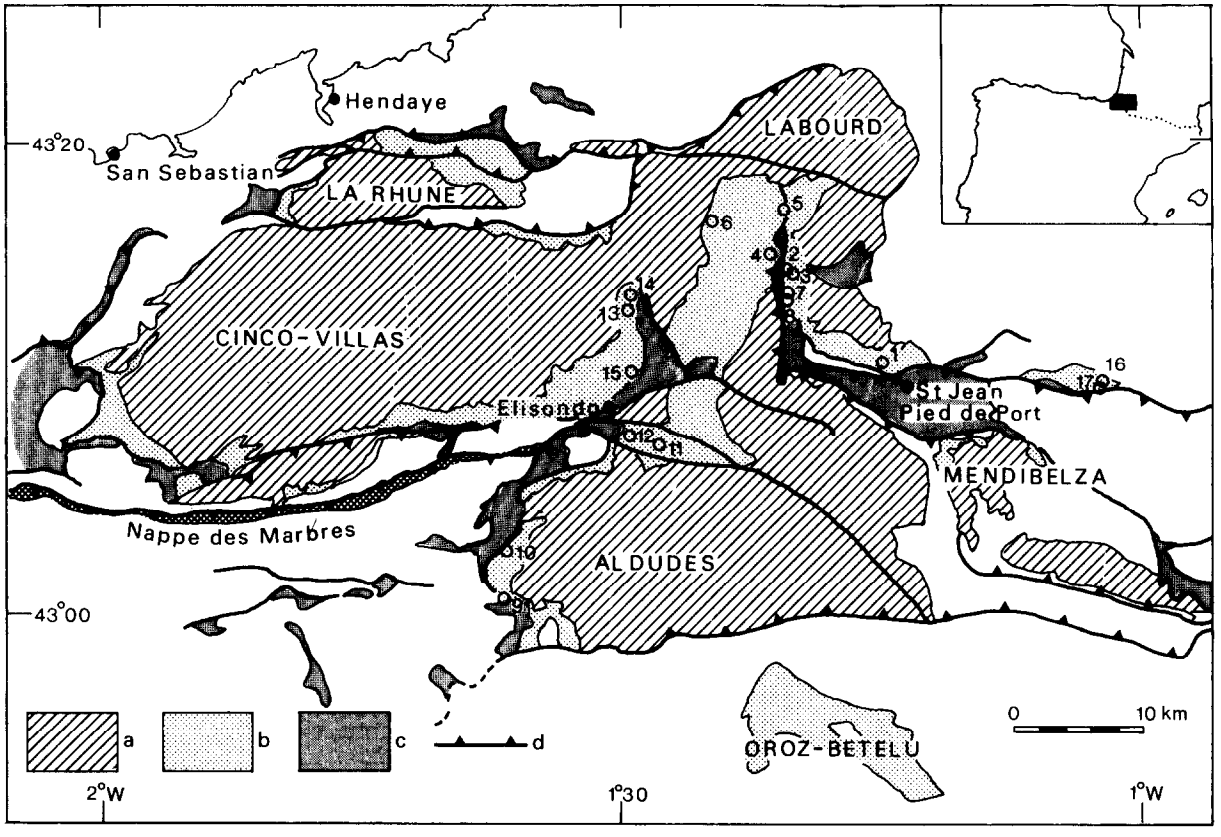


Fig. 1. Geological map of the western Pyrenees and localization of the sampling sites (after Muller and Roger, 1977, simplified). Distribution of the sampling sites over the various tectonic units: *1*—Jara slice; *2*, *3*, *7*, *8*—Baigoura slice; *4*, *5*, *6*—Iparla–Artzamendi slice; *9*, *10*—western border of the Aldudes massif; *11*, *12*—slice north of Aldudes; *13*, *14*, *15*—eastern border of the Cinco-Villas massif; *16*, *17*—north of the Arbailles syncline. *a* = Palcozoic; *b* = Permo-Triassic; *c* = Keuper and Muschelkalk; *d* = major faults and thrusts.

early Triassic, can extend into the late Triassic period. The upper part of the Triassic consists of Keuper argillites and evaporites intruded with tholeiitic bodies (the so-called "ophites"). The magmatic event giving rise to the intrusion of ophites is widespread in the western Pyrenees. But as they are generally located in highly deformed zones, their relationship with the surrounding sediments is unclear. In a few preserved localities, however, they occur as sills interbedded in the Keuper argillites (Walgenwitz, 1976; Beziat, 1983).

2.2. Tectonics

The basic work on the Paleozoic Basque massifs was done by Lamare (1936), who assumed that they were deep-rooted units, like the primary axial zone in the central Pyrenees. More recent studies (Muller and Roger, 1977; Le Pochat, 1982; Hall and Johnson, 1986) suggest a much more intricate structural pattern. According to these authors, the basement consists of two main units (Cinco-Villas–La Rhune, Aldudes), and several thrust slices, the main one being the Labourd massif and the Iparla–Artzamendi and Baigoura–Jara units, (These two last units belong, along with the Jara slice, to the Bidarray basin mentioned in the previous section.) According to Muller and Roger (1977), the present-day structural attitude is the result of three main tectonic events:

(1) A post-Aptian and ante-Cenomanian phase gave rise to subsiding troughs filled in with flyschs, and to a northwards slide of the Jurassic and Cretaceous cover.

(2) The first evident compressive episode took place in the early to middle Eocene. The compression is clearly assessed by the deformations induced in the Mesozoic cover, which behaved independently from the basement, the décollement level being localized in the Keuper argillites and evaporites.

(3) The major phase is dated from the late Eocene. According to Muller and Roger (1977), some Paleozoic slices (Labourd, Baigoura–Jara, Iparla–Artzamendi) underwent clockwise rotations of about 60°, whereas the Cinco-Villas and La Rhune massifs were shifted up to 10 km in the ENE direction. The author's assessments rely upon the measurements of paleocurrents in the Permo-Triassic red beds and of Hercynian structural lineations. Thus, during this period, the Paleozoic basement was broken up into the units and slices mentioned above. They overthrust the Mesozoic cover northwards (Labourd massif, Baigoura–Jara, Iparla–Artzamendi), as well as southwards (Aldudes massif), giving rise to fan-shaped structures (Mattauer, 1968; Choukroune, 1976). The amount of overlapping remains an open question, as pointed out by Le Pochat (1982) and Hall and Johnson (1986). This is of fundamental interest, especially for the Aldudes massif, in connection with the problem of the westward extension of the N.P.F.

The existence of the ante-Cenomanian tectonic event is controversial (Choukroune, 1976; Muller and Roger, 1977; Souquet et al., 1977). This episode may be connected with the so-called North-Pyrenean metamorphism. Its age is still a matter of debate, but a recent extensive K–Ar study (Montigny et al., 1986) has shown strong evidence of ages around 90 Ma.

The instability of the basement relates to late Hercynian faults. The most conspicuous example is the deep-rooted, E–W trending, reverse fault which constitutes the northern front of the "Nappe des Marbres", upthrust towards the north, and abruptly interrupted on the western border of the Aldudes massif. Another typical example is the Elizondo–Pampelona fault running ENE–WSW between the Cinco-Villas and Aldudes massifs.

During the tectonic events summarized above, the Permo-Triassic sequence and the overlying cover behaved quite differently due to the presence of décollement layers localized in the Keuper argillites. The Mesozoic cover has its own style of deformation, characterized by large overthrust folds, whereas the Permo-Triassic coating remained anchored to the basement and underwent folding directly related to the crustal shortening.

Consequently, the Permo-Triassic red beds constitute the most suitable material for paleomagnetic studies related to geodynamical problems. Our sampling sites were located on some of the Paleozoic units and slices (Fig. 1): north of the Arbailles syncline (sites PW16, PW17), the Jara slice (site

PW1), Baigoura slice (sites PW2, PW3, PW7), the Iparla–Artzamendi slice (sites PW4, PW5, PW6), the eastern border of the Cinco-Villas massif (sites PW13, PW15), the western border of the Aldudes massifs (site 10), and a small unit north of the Aldudes massif (site PW12). The sampled beds belong to the Permian (site PW5) or early Triassic (remaining sites). Some dolerite outcrops were tentatively sampled, in order to investigate their suitability for paleomagnetic studies (sites PW8, PW9, PW11, PW14). However, the results obtained on these rocks will be discarded from further considerations owing to uncertainty affecting their tectonic attitude and/or to the complexity of their magnetization.

3. Sampling and analytical techniques

As indicated above, the sampling of the Permo-Triassic rocks was carried out in thirteen localities. Except for localities PW1 and PW5 which consist of coarse sandstone, the Permo-Triassic samples were taken from fine sandstone and silt, light-red to violet–red colored. Splitting parallel to bedding planes could be frequently observed in the siltstone. This made the sampling difficult in some cases.

The sites were hand-sampled or drilled in the field. Between six and ten samples were taken from each site and oriented with magnetic and sun compasses. Two or three standard core specimens, 2.5 cm in diameter and 2.5 cm long, were obtained from each sample.

The NRM of the specimens was measured with a "Digico" fluxgate magnetometer. At least one specimen per sample was submitted to progressive thermal demagnetization consisting of at least eight steps up to 680 °C. The susceptibility was measured initially and after each step, in order to detect the mineralogical changes induced by heat. These effects sometimes gave rise to a noticeable increase in susceptibility and to the acquisition of spurious magnetization which prevented us from achieving a complete demagnetization. In addition, specimens from samples showing complicated thermal behavior were selected for chemical leaching, experimental procedure of which is described elsewhere (Schott and Peres, 1987).

The demagnetization paths were plotted on Zijderveld diagrams and on stereonets. The least squares method described by Halls (1976) and Kirschvink (1980) was used to find the best fit to straight segments on the Zijderveld plots and to great circles on stereonets. The softest components could be generally identified as lines, but the hardest ones were sometimes difficult to isolate. Their direction was then estimated from the end point of the great circle passing through the directions of the two hardest components.

4. Paleomagnetic analysis

The intensities of NRM range from 1.0 to 6×10^{-3} A m^{-1} in sandstone and from 2.0 to 14.0×10^{-3} A m^{-1} in siltstone. The scattering of directions is very dependent on lithology: in coarse-grained beds it is higher than in fine-grained beds. In siltstone, directions of NRM display a streaked distribution, thus revealing the presence of two magnetic components.

Stepwise thermal demagnetization generally allows the determination of two components, both possessing high blocking temperatures. The removal of the first one requires cleaning steps up to 670 °C, the hardest one being isolated in the interval bounded by the Curie temperature of haematite. This high-temperature component is always interpreted as the primary magnetization having a late Permian–early Triassic age (throughout the rest of the paper late Permian–early Triassic will be referred to as Permo-Triassic). This assumption is based on the inclinations which are, after tilting correction, consistent with the expected Permo-Triassic values in this area.

Except for site PW4, in sandstone (sites PW1, PW2, PW3, PW4, PW5, PW6, PW7, PW12), the second component is considered either as Permo-Triassic direction or as an unresolved intermediate direction between antiparallel Permo-Triassic components. The assumption of superimposition of antiparallel components relies on the observation of both polarities made in several sites (PW3, PW6, PW7). In some favorable instances, antiparallel components could be determined in the same specimen (Fig. 2), but generally it was impossible to calculate the two directions accurately.

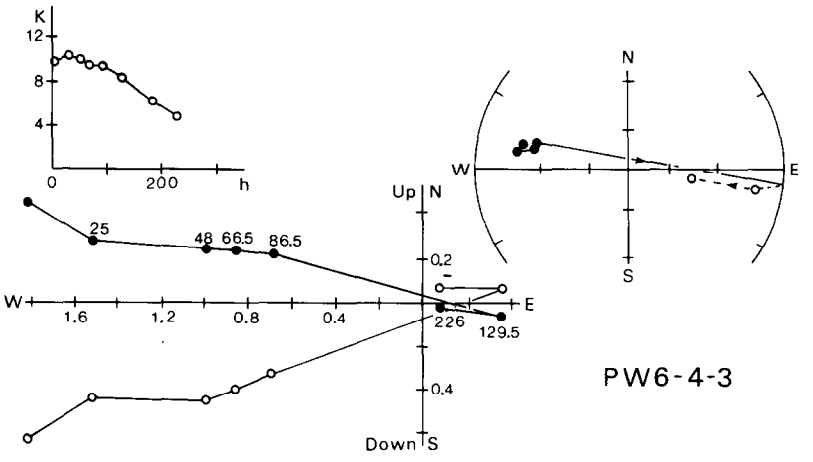


Fig. 2. Orthogonal and stereographic plot showing the superposition of two nearly opposite magnetic components (chemical demagnetization). On the orthogonal diagrams, solid circles and projections onto the horizontal plane, open circles are projections onto the vertical, E–W plane. Units along the axes are in 10^{-3} A/m. On the Wulff-net plot, solid circles are projections onto the lower hemisphere. *K*—susceptibility per unit volume in 10^{-5} S.I.; *h*—cumulative hours of leaching.

The superimposition of antiparallel components is a common situation in red beds (for instance, Larson et al., 1982; Turner, 1981).

Contrary to coarser-grained levels, in siltstones (sites PW10, PW13, PW15, PW16, PW17), only the hardest component is assumed to reflect the Permo-Triassic magnetic field. But its isolation is often obscured by the occurrence of mineralogical transformations induced by heat above 600 °C. Unfortunately, better separation of the components is not possible by chemical cleaning, due to the overlapping of stability spectra. This inconvenience is illustrated by Fig. 3. Figure 3a displays the Zijderveld diagram given by thermal cleaning. It clearly shows the presence of two components (without taking into account a soft component removed at 270 °C). Figure 3b is the Zijderveld diagram of combined chemical and thermal demagnetization. In the first part of the diagram, corresponding to the acid leaching, the magnetization behaves practically like a single component, with an intermediate direction between those of

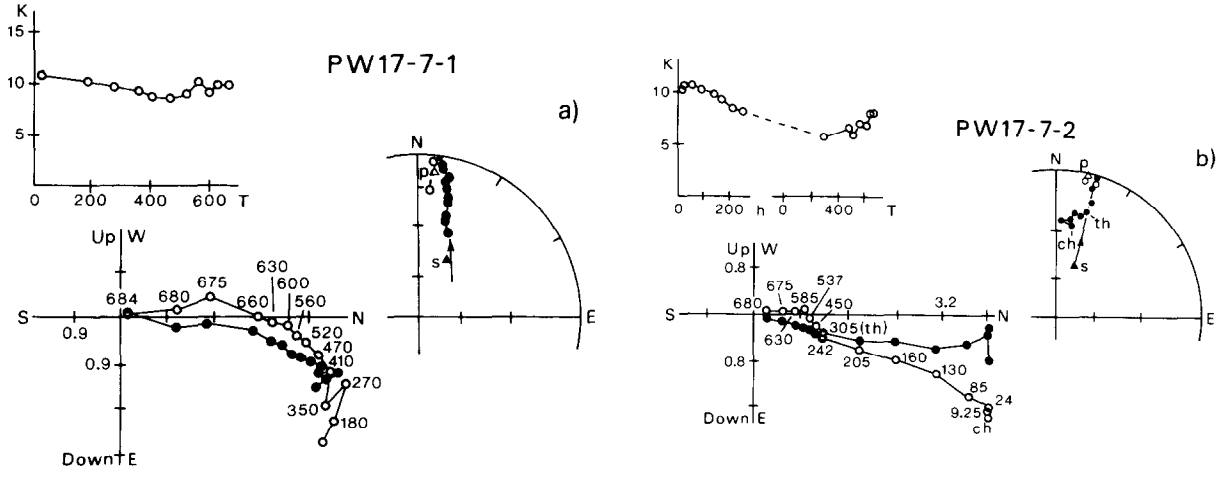


Fig. 3. Comparison of (a) thermal and (b) combined chemical and thermal demagnetizations (the chemical treatment was stopped after 242.5 h cumulative leaching). On the orthogonal diagrams, open circles are projections onto the vertical, N–S plane. *P*—primary, *S*—secondary component; *ch*—chemical leaching; *th*—thermal demagnetization. Other symbols as in Fig. 2.

the thermally defined components. But subsequent thermal cleaning again reveals that two components are actually present with directions consistent with the results of the thermal analysis.

The significance of the secondary components identified in siltstone will be discussed in the next section.

5. Discussion of the results

5.1. Primary directions

The site mean directions of components interpreted as representing the original, late Permian to early Triassic, directions are displayed on Fig. 4 and Table 1 (site PW16 having only given two primary directions, sites PW16 and PW17 were grouped into one single site PW17. In this case, within-site statistics were calculated after tilting correction, otherwise they have the same values before and after correction.)

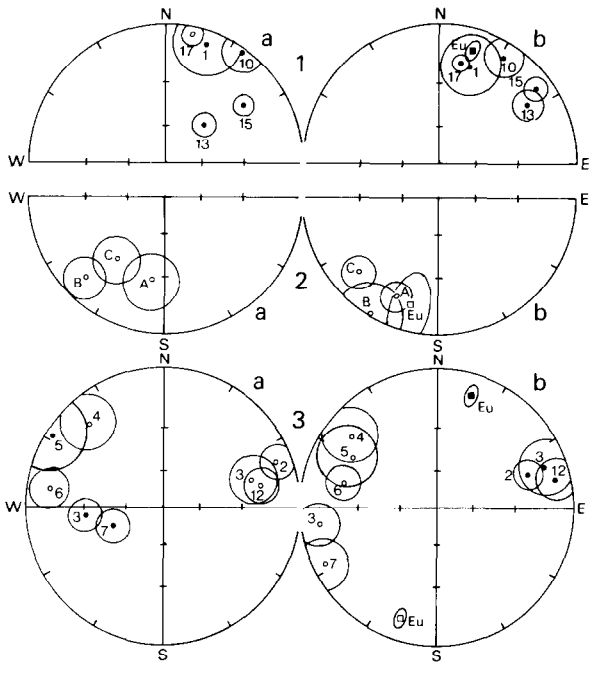


Fig. 4. Stereographic plot of the site mean directions and circles of confidence for the Permian or Permo-Triassic components.

1. Jara slice (*1*), North of Arbailles (*17*), western border of Aldudes (*10*), eastern Border of Cinco-Villas (*13, 15*).

2. Permian basalt flows (after Van der Voo and Boessenkool, 1973). *A*—La Rhune; *B*—northern border of Cinco-Villas; *C*—southern border of Cinco-Villas.

3. Baigoura (*2, 3, 7*), Iparla–Artzamendi (*4, 5, 6*) and north of Aldudes (*12*) slices.

a—before tectonic corrections; b—after tectonic correction. *Eu*—European reference directions (after Westphal et al., 1986). Solid circles are projections onto the lower hemisphere.

The results are divided into two groups, depending on the declinations obtained after tilting corrections. The first group consist of directions which are either close to that characterizing stable Europe for the same period (north of the Arbailles syncline, Jara slice), or which are shifted slightly clockwise (eastern border of the Cinco-Villas massif, western border of the Aldudes massif). The second group displays extreme clockwise rotations relative to the European direction (Baigoura and Iparla–Artzamendi slices, small unit north of the Aldudes massif). All the results are readily interpreted as a consequence of dextral tectonic rotations of various amplitudes, as summarized in Table 2. In addition to our results, previous data are also shown in Fig. 4 and Table 2. Original values from Van der Voo and Boessenkool (1973) are grouped according to their geographical distribution and mean directions calculated for each of the three groups (La Rhune massif, northern and southern borders of the Cinco-Villas massif). Table 2 shows that our results are consistent with the data yielded by the upper Permian basaltic flows. The red beds from the Oroz-Betelu massif were investigated by Van der Voo (1979). The characteristic directions obtained after thermal or a.f. demagnetization were quite scattered and the author used density distribution in order to find out a mean direction. We tried to improve the location of the mean by Van Alstine's method (1980). It allowed the calculation of a normal and reverse mean direction, located respectively at $D = 348.5°$, $I = 20.0°$ and $D = 170.0°$, $I = -10.0°$. In Table 2 angles of rotation are defined as a tectonic unit mean declination minus reference declination. For all units, reference is the European early Triassic direction, except for Oroz–Betelu where the Iberian direction was used. In this way, clockwise rotations can be inferred everywhere. Further reasons for the use of different references will be discussed in section 6.

TABLE 1

Site mean directions and paleomagnetic poles of Permo-Triassic magnetization

Site	Polarity	n/n$_o$	Site mean directions				Statistics			Paleomagnetic poles			
			without tect.cor.		with tect.cor.								
			D_m	I_m	D_m	I_m	N	k	α_{95}	Lat.N	Long.E	dp	dm
PW1	N	16/ 6	19.5	6	18	18	16	4	16.0	53	149	8.5	16.5
PW2	N	14/ 6	68.5	- 7	70	19.5	13	27	8.0	21	95	4.5	8.0
PW3	N	16/ 8	73	-22	69.5	10	7	20	13.5	18	99	7.0	14.0
	R		263.5	31	261.5	- 8.5	4	76	10.5	9	91	5.5	10.5
PW4	R	15/ 8	317.5	-12.5	309.5	-12.5	4	41	14.0	22	235	7.5	14.5
PW5	R	16/ 8	302	3	301	-18.4	14	8	15.5	15	240	8.5	16.0
PW6	M	17/ 8	279	-10.5	284.5	-20	7	42	9.5	3	251	5.0	9.5
PW7	M	17/ 9	248.5	47.5	243	- 5.5	5	41	12.0	21	106	6.0	12.0
PW10	M	14/ 9	35.5	2	32	6.5	7	43	9.0	41	134	4.5	9.0
PW12	N	17/ 9	77.5	-18	76.5	6.5	10	26	9.5	12	95	4.5	9.5
PW13	N	17/11	46.5	47.5	57	15.0	7	55	8.0	29	106	4.0	8.5
PW15	N	12/ 9	54.5	20	53	7	7	110	5.5	29	113	3.0	5.5
PW17*	N	11/ 6	12	- 3.5	12.5	17	8	155	4.5	54	158	2.5	4.5

Polarity: N—normal; M—mixed; R—reversed. n/no—number of specimens/number of samples. N—number of directions used in the Fisher statistics. k, α_{95}—parameter of Fisher statistics. dp, dm—semi-angles of the ellipse of confidence. Paleomagnetic poles are computed after tectonic corrections.

* site PW16 having yielded only two primary directions sites PW16 and PW17, which are only 150 m apart, were grouped into one single site PW17.

5.2. *Secondary directions*

These components are identified mostly in the fine-grained beds. The fold test shows clearly that the magnetization postdates the folding (Fig. 5). But the in-field directions display a suggestive distribution: the declinations are scattered whereas the inclinations are well-grouped between 40° and 50°. Taking into account the tectonic implication of the primary directions, we suggest that the distribution of directions observed results from two successive tectonic phases: (1) folding with subhorizontal axes and (2) clockwise rotations around the vertical axes. In this simplified sketch the secondary magnetization would have been built up after the first phase. The examination of the consequence of the second phase upon the secondary components is slightly hampered by the uncertainty affecting their age and, therefore, the direction of the paleomagnetic field in which they were built up. The inclinations give some information about the age of the remagnetization. Rather low values, around 40°, are found in various localities belonging to the Tethysian domain, as pointed out by Westphal et al. (1986). In the Pyrenean range, directions with similar inclina-

TABLE 2

Mean Permo-Triassic directions for the different tectonic units and apparent rotations relative to stable Europe *

Unit	Site mean directions				Statistics					Angle of rotation relative to stable Europe
	without tect.cor.		with tect.cor.		without cor.			with cor.		
	D_m	I_m	D_m	I_m	N	k	95	k	α_{95}	
N. of Arbailles (sites PW16, PW17)	12	- 3.5	12.5	17	1	-	-	-	-	- 4.5 ± 5.0
Jara slice (site PW1)	19.5	6	18	18	1	-	-	-	-	1.0 ± 16.5
(1) La Rhune	188	-27.5	203	-14.5	4	30	17	126	8.0	8.5 ± 10.0
(1) Cinco-Villas (N)	216.5	-31	210.5	- 2	4	41	14.5	41	14.5	16.0 ± 13.5
(1) Cinco-Villas (S)	224	-11.5	227	-13.5	9	24	11	38	8.5	32.5 ± 10.0
Cinco-Villas (E) (sites PW13, PW15)	51.5	34	55	11	2	-	-	-	-	38.0 ± 4.5
Aldudes (W) (site PW10)	35.5	2	32	6.5	1	-	-	-	-	15.0 ± 9.5
N. of Aldudes (site PW12)	77.5	-18	76.5	6.5	1	-	-	-	-	59.5 ± 10.0
Baïgoura (sites PW2, PW3, PW7)	74.5	-27	72	11	4	21	20.5	81	10.5	55.0 ± 8.5
Iparla-Artza. (sites PW4, PW5, PW6)	299.5	- 7	298.5	-17	3	15	33	40	19.5	102.5 ± 15.0
										Angle of rotation relative to Iberia
(2) Oroz-Betelu	-	-	348.5	20	-	-	-	65	4.0	13.5 ± 5.5

* Statistical parameters calculated either on the site-mean directions (when more than two sites per unit) or on sample directions (when only one or two sites per unit). European reference pole is from Westphal et al. (1986). Errors on angles of rotation are computed using the formulae and corrections given by Demarest (1983). (1)—mean unit directions recalculated from the original data of Van der Voo and Boessenkool (1973) by grouping site mean directions in accordance with their geographical distribution; (2)—mean normal direction recalculated using the original data of Van der Voo (1969).

tions are clearly post-tectonic everywhere (Schott, 1985; Schott and Peres, 1987), and the same values were also obtained by Roggenthen (1976) in Paleocene sediments along the coast of the Bay of Biscay. Taking into account the chronology of tectonic events outlined in section 2.2, we feel that a late Eocene–Oligocene age is a reasonable assumption. Assuming a value of 5° deduced from the Paleocene–Eocene paleomagnetic pole of stable Europe (Westphal et al., 1986), Table 3 shows

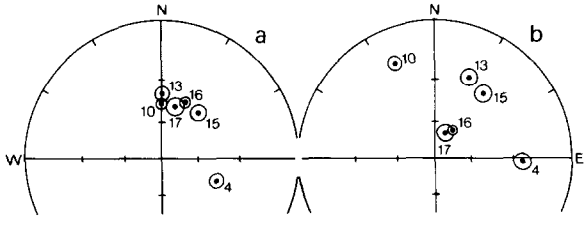


Fig. 5. Stereographic plot of the site mean directions and circles of confidence for the secondary magnetization in red beds. a. Before tectonic corrections. b. After tectonic corrections. Symbols as in Fig. 4.

TABLE 3

Comparison of angles of rotation of various tectonic units deduced from secondary directions with values inferred from Permo-Triassic directions *

Site	Site mean directions				Statistics			Angle of rotation	
	without tect.cor.		with tect.cor.						
	D_m	I_m	D_m	I_m	N	k	α_{95}	observed	expected
PW4	113	43	92.5	24.5	6	160	5.0	107.5 ± 8.0	102.5 ± 15.0
PW10	359.5	46.5	337.5	16	8	240	3.5	- 6.0 ± 7.0	15.0 ± 9.5
PW13	6.5	41.5	28	23.5	16	146	4.0	1.0 ± 7.5	38.0 ± 4.5
PW15	39.5	43.5	37	29	7	125	5.5	34.0 ± 8.0	38.0 ± 4.5
PW16	22.5	41.5	32.5	62.5	23	100	4.0	17.0 ± 7.5	-4.5 ± 5.0
PW17	15	47.5	22	67.5	10	53	6.5	9.5 ± 9.0	-4.5 ± 5.0

* European reference poles are those listed by Westphal et al. (1986). (Paleocene–Eocene).

that the rotations deduced from the secondary components agree with the values based on the primary directions in sites PW4, PW15, PW16, PW17.

The secondary mean direction of site PW13, on the other hand, is not rotated and could be interpreted as post-dating the rotations. Thus the secondary directions would be penecontemporaneous with the second tectonic phase.

6. Tectonic implications

Figure 6 displays the distribution of angles of rotation inferred from paleomagnetic data throughout the area studied, along with some other results labelled U9, U10, U11, U12. Directions U10, U11 are given by Vandenberg (1979), direction U12 by Roggenthen (1976). Direction U10 comes from the study of Triassic–Jurassic limestone and shale, direction U11 from Albian flysch and direction U12 from Paleocene pelagic limestone. In these three studies the formations are located above décollement levels (late Triassic and/or Urgonian), therefore one must be careful in their interpretation. We note, however, that the directions are similar to stable Europe directions (U11, U12) or shifted clockwise (U10).

We will now discuss the hypothesis which may explain this pattern of declinations and hence of rotations. It is well known that two major events are responsible for the structures observed in the Pyrenean range: the opening of the Bay of Biscay dated from the late early–early late Cretaceous and the convergence of the European and Iberian plates beginning in the late Cretaceous. Within this general framework three hypotheses can be envisaged:

(a) The entire western Pyrenees were located in an intermediate zone during the mid-Cretaceous displacement of Iberia. The scattering of declinations would have resulted from local movements occurring during that shift.

(b) The entire western Pyrenees belonged to the same plate by the time the Bay opened and the scattering of declinations is a local effect induced by the subsequent early Tertiary compression. Under this assumption, we note the dissimilar behavior among the Cinco-Villas and the Aldudes massifs, the small slices east and north of them (that is, generally speaking, the area situated north of

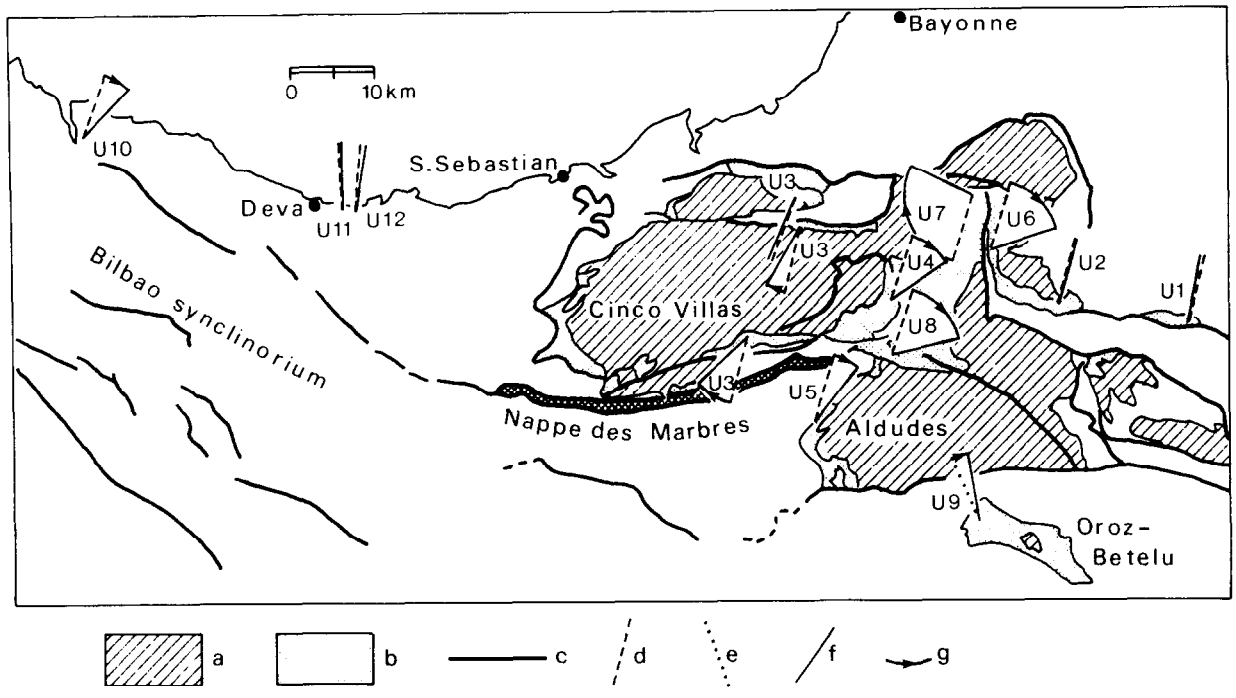


Fig. 6. Paleomagnetic declinations measured throughout the western Pyrenean range in rocks of various ages and comparison with the contemporaneous European (all but U9) or Iberian (U9) reference declinations. Paleomagnetic data are from: U3—Van der Voo and Boessenkool (1973), late Permian; U9—Van der Voo (1969), Triassic; U10—VandenBerg (1979), late Triassic, early Jurassic; U11—VandenBerg (1979), Albian; U12—Roggenthen (1976), Paleocene; other localities: this study. *a*—Paleozoic; *b*—Permo-Triassic; *c*—main faults and thrusts; *d*—European reference directions; *e*—Iberian reference direction; *f*—localities mean directions; *g*—way of rotation.

the "Nappe des Marbres"), on the one hand, and the Oroz-Betelu massif on the other: if the whole area belonged to the European plate we observe clockwise rotations in the first part and a counter-clockwise rotation in the second; if, to the contrary, the whole area was part of Iberia, the amount of clockwise rotation changes abruptly between the Oroz-Betelu massif and the other massifs and slices.

(c) The Oroz-Betelu massif belonged to Iberia and the other units to the European plate during the mid-Cretaceous motion. In this case, as in case (b), the pattern of declinations observed at present is due to the early Tertiary compression. Clockwise rotations occurred everywhere.

The test of the above mentioned hypotheses is closely related to the kinematic model which is assumed to describe the motion of the Iberian plate with respect to Europe. There are, generally speaking, two types of models which are consistent with paleomagnetic data. We will set out the discussion in light of both theories, which will be referred to as single rotation models and left lateral shift models respectively.

6.1. Single rotation models

This category of model, represented in its most recent version by the work of Masson and Miles (1984), assumes that during the main phase of evolution, contemporary with the opening of the Bay of Biscay, the Iberian plate rotated without major transcurrent motion in the Pyrenees. This evolution is advocated by authors who deny any prominent role to the North Pyrenean Fault (Souquet and Mediavilla, 1976; Souquet et al., 1977; Souquet and Debroas, 1980). It is rather difficult to look for a boundary under this assumption, because it implies the existence, prior to the rotation, of a domain of collision between the plates whose width increased eastwards, and which experienced a very large amount of compression

and shortening. Bearing in mind the paleomagnetic data obtained from the southern flank of the axial zone (Van der Lingen, 1960; Schwarz, 1963; Van Dongen, 1967; Cogné, 1983), we can only claim that the intermediate domain is located north of this E–W trending zone. Any paleomagnetic direction can be expected in the intermediate domain, so that the test of assumptions (a) (b) and (c) is quite inconclusive and of little geological interest because they never involve any significant differential motion between the different basement units building the western Pyrenees.

6.2. Left-lateral shift models

This second kind of model, first proposed by Le Pichon et al. (1971), and whose most elaborate version was published by Olivet et al. (1984), involves a major left-lateral shift of the Iberian plate during the middle Cretaceous. Although the transform zone is probably broader than suggested in the original simplified model, the concept of a boundary associated with this movement is obvious.

Hypothesis (a) assumes that the whole western Pyrenees were included in the transform zone whose width would have reached at least 60 km. Such a large shear zone was suggested by several authors (Choukroune and Mattauer, 1978; Bessière and Soula, 1978) but the geological and geophysical evidence supporting this assumption is weak.

Hypotheses (b) and (c) are more plausible. But here, their geological implications are of major importance because in the initial position of the plates, it would have been necessary for the parts belonging to the Iberian plate to have been displaced some 400 km to the west relative to the European plate. As there is strong geological evidence supporting the location of the Oroz-Betelu massif on the Iberian margin and the Cinco-Villas massif on the European margin, hypothesis (c) fits the known constraints better. However, since there are some geological interpretations locating the Aldudes massif on the Iberian margin (Souquet et al., 1977), this unit deserves some further discussion. We feel that the Aldudes massif belonged to the European plate during the mid-Cretaceous shift for the following reasons:

(1) With respect to the European early Triassic direction, the rotation of the site mean direction is quite similar to the values found in the Cinco-Villas massif. With respect to the Iberian reference direction, the rotation would amount to 50°, a value characteristic of the small thrust slices.

(2) If the Aldudes massif were attached to the Iberian plate, it would have had to be moved 400 km westwards prior to the initial movement, as noted above. This implication can hardly be accepted on geological grounds.

Thus, we suggest that during the mid-Cretaceous movement of Iberia which led to the opening of the Bay of Biscay, and which is the period allowing the clearest definition of a boundary between Iberia and Europe, this boundary ran, in the western Pyrenees, south of the Cinco-Villas and Aldudes massif.

The rather simple ordering of the units composing the western Pyrenees was markedly disturbed by the early Tertiary compression whose effects are described by Muller and Roger (1977) and Hall and Johnson (1986). To continue with hypothesis (c), the clockwise rotations inferred from the distribution of Permo-Triassic declinations are attributed to the local effects of the compression. We notice that, under assumption (c), both types of model lead to the same interpretation for this period. This is not surprising because from the magnetic anomaly 34 onwards, the models of Masson and Miles and Olivet et al. are identical. Both models assume a convergence with a dextral transcurrent movement in the Pyrenean range which is consistent with the clockwise rotations observed in the area studied. To the south of the Cinco-Villas massif, the deep-rooted E–W trending zone which represents the northern front of the "Nappe des Marbres" could be the trace of the fossil transform zone acting during the mid-Cretaceous, as suggested by several authors (Van der Voo and Boessenkool, 1973; Rat et al., 1983). This feature is abruptly interrupted on the western border of the Aldudes massif, a fact which could be accounted for by a southward displacement of the Aldudes massif (Rat, 1983; Le Pochat, 1982), overthrusting the Iberian margin in its present-day position. Therefore, in depth, the mid-Cretaceous boundary may be located beneath the Aldudes

massif. The idea of a North Pyrenean Fault cut by younger reversed faults and covered by thrust sheets is also advocated by Hall and Johnson (1986). Some geophysical data support our interpretation. From a seismic profile running through the Cinco-Villas and Aldudes massifs, Daignières et al. (1982) inferred a discontinuity of the depth of the Moho which passes from 40 km to 30 km in and E–W direction. They located the step beneath the "nappe des marbres" (we note that the location also coincides with the Elizondo–Pampelona fault). The crustal thinning under the Cinco-Villas massif is to be compared with similar behaviour observed in the central Pyrenees between the axial zone and the North-Pyrenean zone. The fact that no thinning occurs beneath the Aldudes massif could be explained by the overthrust of this unit upon the Iberian margin. Lastly, let us also mention the interpretation of a geomagnetic sounding carried out in the western Pyrenees by Galdeano et al. (1980), who assumed the existence of a deep-rooted accident running south of the Aldudes massif (but rather unprecisely located) and stretching eastwards along the "Bigorre fault" (Souquet et al., 1977).

7. Conclusions

The paleomagnetic investigations carried out on upper Permian–lower Triassic red beds show that major clockwise rotations occurred in the different structural units recognized in the western Pyrenees. Taking into account primary (Permo-Triassic) as well as secondary (late Eocene ?) directions, we suggest that this pattern of clockwise rotations displaying a large range of values (0–102°) results from the early Tertiary compressional phase. However, in a model involving a single rotation of the Iberian plate, the concept of a boundary between the European and Iberian plates is fuzzy, and local rotations can occur at any time during the movement. On the other hand, things are clearer in a model advocating a left-lateral shift of Iberia inducing a transcurrent displacement in the Pyrenees. Under this assumption, we believe that, during the mid-Cretaceous opening of the Bay of Biscay, the boundary between the European and Iberian plates ran between the Cinco-Villas–Aldudes massifs and the Oroz-Betelu massif. North of the limit, directions were similar to the stable Europe reference, whereas areas located south of the limit displayed Iberian-like directions. As suggested above, this rather simple ordering was disturbed by the subsequent convergence of the plates. Whatever the validity of our interpretation, the paleomagnetic results emphasize the quite complicated structure of the western Pyrenees, which are split into numerous units and slices having experienced strong deformations. Considering the intricacy of this area, further investigations are needed in order to give a more detailed and documented distribution of the paleomagnetic directions. For reasons exposed in section 1, Permo-Triassic red beds are the most suitable material. Unfortunately, they have undergone strong remagnetizations which require careful analysis. However, secondary magnetizations can also yield useful information, as suggested by our interpretation.

Acknowledgements

The authors are indebted to C. Lucas for his valuable suggestions for the field work and improvement of the geological setting, and to J. Muller for helpful discussions. M. Westphal provided helpful comments on the manuscript. Ms. L. Khan kindly improved the English version of the text. The work was supported by the Institut National des Sciences de l'Univers through the national research project ATP géodynamique. Publication of this article has been authorized by the Comision Nacional de Geologica in Madrid; their cooperation is gratefully acknowledged. This is a contribution of CNRS UA 323.

References

Bessière, G. and Soula, J.C., 1978. La formation des Pyrénées: exemple d'évolution d'une zone de cisaillement complexe. 6e Réun. Annu. Sci. Terre, Paris, p. 36.

Beziat, D., 1983. Etude pétrologique et géochimique des ophites des Pyrénées: implications géodynamiques. Thèse 3e cycle, Univ. Toulouse III, 60 pp.

Bixel, F. and Lucas, C., 1983. Magmatisme, tectonique, et sédimentation dans les fossés stéphano-permiens des Pyrénées occidentales. Rev. Géogr. Phys. Géol. Dyn., 24: 329–342.

Boillot, G. and Capdevilla, R., 1977. The Pyrenees: subduction and collision? Earth Planet. Sci. Lett., 35: 151–160.

Choukroune, P., 1976. Structure et évolution tectonique de la zone nord-pyrénéenne. Analyse de la déformation dans une portion de la chaîne à schistosité sub-verticale. Mém. Soc. Géol. Fr., LV, 127: 1–116.

Choukroune, P. and Mattauer, M., 1978. Tectonique des plaques et Pyrénées: sur le fonctionnement de la faille transformante nordpyrénéenne; comparaison avec des modèles actuels. Bull. Soc. Géol. Fr., (5), 20: 689–700.

Cogné, J.P., 1983. Etude paléomagnétique et déformation inverse de l'aimantation de séries naturelles déformées. Les schistes rouges des Alpes Maritimes et du col du Somport (Pyrénées occidentales). Thèse 3e cycle, Univ. Rennes, 108 pp.

Daignières, M., Gallart, J., Banda, E. and Hirn, A., 1982. Implications of the seismic structure for the orogenic evolution of the Pyrenean Range. Earth Planet. Sci. Lett., 57: 88–100.

Demarest, H.H., 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. J. Geophys. Res., 88: 4321–4328.

Doubinger, J. and Mouline, M., 1961. Sur la flore fossile du Carbonifère de Larroun et de l'Ibantelly (Basses Pyrénées). C.R. Acad. Sci. Paris, 253: 3029–3031.

Galdeano, A., Menvielle, M., Merino del Rio, J., Rossignol, J.C. and Tessier, C., 1980. Sur l'existence d'accidents transversaux à la chaîne pyrénéenne: apport des sondages géomagnétiques profonds dans le Pays Basque. C.R. Acad. Sci. Paris, 290: 227–230.

Hall, C.A. and Johnson, J.A., 1986. Apparent western termination of the North Pyrenean Fault and tectonostratigraphic units of the western north Pyrenees, France and Spain. Tectonics, 5: 607–627.

Halls, H.C., 1976. A least-squares method to find a remanence direction from converging remagnetization circles. Geophys. J. R. Astron. Soc., 45: 297–304.

Herbert, M., 1881. Le terrain pénéen de la Rhune et l'étage coral lien des Pyrénées. Bull. Soc. Géol. Fr., (3), 9: 179–181.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data. Geophys. J. R. Astron. Soc., 62: 699–718.

Lamare, P., 1936. Recherches géologiques dans les Pyrénées Basques d'Espagne. Mém. Soc. Géol. Fr., 12 (27): 469 pp.

Larson, E.E., Walker, R., Patterson, P.E., Hoblit, R.P. and Rosenbaum J.G., 1982. Paleomagnetism of the Moenkopi formation, Colorado Plateau: basis for long term model of acquisition of chemical remanent magnetism in red beds. J. Geophys. Res., 87: 1081–1106.

Le Pichon, X., Bonnin, J., Francheteau, J. and Sibuet, J.C., 1971. Une hypothèse d'évolution structurale du Golfe de Gascogne. In: J. Debyser, X. Le Pichon and L. Montadert (Editors), Histoire Structural du Golfe de Gascogne, Technip, VI (11): 1–44.

Le Pochat, G., 1982. Reconnaissance des écailles de cristallin et de Paléozoique dans les massifs paléozoiques basques. Progr. Géol. Prof. Fr., Bur. Rech. Géol. Min., 7.1: 285–287.

Lucas, C., 1977. Le Trias des Pyrénées, corrélations stratigraphiques et paléogéographie. Bull. Bur. Rech. Géol. Min., 4 (3): 225–231.

Lucas, C., 1985. Le grès rouge du versant nord des Pyrénées. Essai sur la géodynamique de dépots continentaux du Permien et du Trias. Thèse Doct. Etat, Univ. Toulouse, 267 pp.

Lucas, C., Doubinger, J. and Broutin, J., 1980. Premières datations palynologiques dans les grès triasiques des Pyrénées. C.R. Acad. Sci. Paris, 291: 517–520.

Masson, D.G. and Miles, P.R., 1984. Mesozoic sea-floor spreading between Iberia, Europe and North America. Mar. Geol., 56: 279–287.

Mattauer, M., 1968. Les traits structuraux essentiels de la chaîne pyrénéenne. Rev. Géogr. Phys. Géol. Dyn., 10 (1): 3–11.

Montigny, R., Azambre, B., Rossy, M. and Thuizat, R., 1986. K–Ar study of Cretaceous magnetism and metamorphism in the Pyrenees: age and length of rotation of the Iberian Peninsula. Tectonophysics, 129: 257–273.

Muller, J. and Roger, Ph., 1977. L'évolution structurale des Pyrénées (domaine central et occidental). Le segment hercynien, la chaîne de fond alpine. Géol. Alp., 53 (2): 149–192.

Olivet, J.L., Bonnin, J., Beuzart, P. and Auzende, J.M., 1984. Cinématique de l'Atlantique Nord et Central. C.N.E.X.O., Rapp. Sci. Techn., 54: 108 pp.

Rat, P., 1983. Les régions basco-cantabriques et nord-ibériques: présentation, problèmes posés. Mém. Géol. Univ. Dijon, 9: 1–24.

Rat, P., Amiot, M., Feuillée, P., Floquet, M., Mathey, B., Pascal, A., and Salomon, J., 1983. Vue sur le Crétacé basco-cantabrique et nord ibérique, conclusions, remarques, suggestions. Mém. Géol. Univ. Dijon, 9: 177–180.

Roggenthen, W.M., 1976. Magnetic stratigraphy of the Paleocene. A comparison between Spain and Italy. Mém. Soc. Géol. Ital., 15: 73–82.

Schott, J.J., 1985. Paléomagnétisme des séries rouges du Permien, du Trias et du Crétacé inférieur dans les Chaînes Pyrénéo-Cantabrique et NW-Ibérique. Implications Géodynamiques. Thesis, Strasbourg, 382 pp.

Schott, J.J. and Peres, A., 1987. Paleomagnetism of the lower Cretaceous red beds from northern Spain: evidence for a multi-stage acquisition of magnetization. Tectonophysics, 139: 239–253.

Schwarz, E.J., 1963. A paleomagnetic investigation of Permo-Triassic redbeds and andesites from the spanish Pyrenees. J. Geophys. Res., 68: 3265–3271.

Soula, J.C., Lucas, C. and Bessière, G., 1979. Genesis and evolution of Permian and Triassic basins in the Pyrenees by simple shear acting on old Variscan structures: field evidence and experimental models. Tectonophysics, 58: T1–T9.

Souquet, P. and Debroas, E.J., 1980. Tectonogenèse et évolution des bassins de sédimentation dans le cycle alpin des Pyrénées. Mém. Bur. Rech. Géol. Min., 107: 213–233.

Souquet, P. and Mediavilla, F., 1976. Nouvelle hypothèse sur

la formation des Pyrénées. C.R. Acad. Sci. Paris, 282: 2139–2142.

Souquet, P., Peybernes, B., Billote, M. and Debroas, E.J., 1977. La chaine alpine des Pyrénées. Géol. Alp., 53 (2): 193–216.

Turner, P., 1981. Relationship between magnetic components and diagenetic features in reddened Triassic alluvium (St. Bees Sandstones, Cumbria, UK). Geophys. J. R. Astron. Soc., 67: 395–413.

Van Alstine, D.R., 1980. Analysis of the modes of directional data with particular reference to palaeomagnetism. Geophys. J. R. Astron. Soc., 61: 101–113.

VandenBerg, J., 1979. Paleomagnetism and the changing configuration of the western Mediterranean area in the Mesozoic and early Cenozoic eras. Geol. Ultraiectina, 179 pp.

Van der Lingen, G.J., 1960. Geology of the Spanish Pyrenees, North of Canfranc, Huesca province. Estud. Geol., Madrid, 16: 205–242.

Van der Voo, R., 1969. Paleomagnetic evidence for the rotation of the Iberian Peninsula. Tectonophysics, 7: 5–56.

Van der Voo, R. and Boessenkool, A., 1973. Permian paleomagnetic result from the western Pyrenees delineating the plate boundary between the Iberian Peninsula and stable Europe. J. Geophys. Res. 78: 5118–5127.

Van Dongen, P.G., 1967. The rotation of Spain: paleomagnetic evidence from the eastern Pyrenees. Paleogeogr., Paleoclimatol., Paleoecol., 3: 417–432.

Walgenwitz, F., 1976. Etude pétrographique des roches intrusives triasiques, des écailles du socle profond et des gites de chlorite de la région d'Elizondo (Navarre espagnole). Thèse 3e cycle, Univ. Besancon, 167 pp.

Westphal, M., Bazhenov, M.L., Lauer, J.P., Pechersky, D.M. and Sibuet, J.C., 1986. Paleomagnetic implications on the evolution of the Tethys Belt from the Atlantic Ocean to the Pamirs since the Triassic. Tectonophysics, 123: 37–82.